

Scalability & Future Plan

Date	13 July 2026
Team ID	XXXXXX
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports limited concurrent users	Performance may decrease with heavy traffic	High
2	Stores data locally using JSON files	Not suitable for large-scale data storage	Medium
3	Basic authentication and security	Needs stronger protection for production	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports small number of users	Deploy using cloud services with load balancing
2	Data Storage	Local JSON storage	Migrate to MySQL/PostgreSQL database
3	Performance	Single FastAPI server	Add caching and asynchronous processing
4	Security	Basic API validation	Implement JWT authentication and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User Login & Profile Management	1–2 Months	Personalized learning experience
Phase 2	Multi-language Support	2–3 Months	Increased accessibility
Phase 3	Mobile Application	3–6 Months	Better user reach and convenience
Phase 4	AI Progress Tracking & Analytics Dashboard	6+ Months	Advanced learning insights and recommendations